

Document Title: 1.2 Problem Statement Context

Author: Santhosh Akkiseti

Project: CampusAssist AI

Phase: 1. IDEATION PHASE

1.2 Problem Statement Context

1. Background

Modern educational institutions operate heavily on digital infrastructure. Students rely on university networks, hardware resources, learning management systems, and centralized institutional accounts for their daily academic activities. A seamless, high-availability IT support framework is essential to ensure that technical disruptions do not impede learning or administration.

2. Current Campus IT Support Situation

In many university environments, IT support is managed through traditional, manual channels such as central email inboxes, physical helpdesk centers, or basic web-based submission forms. When students encounter technical difficulties—such as WiFi disconnects during exams, software configuration errors, or account lockouts—they must navigate disjointed support channels that often lack real-time assistance or guided self-service capabilities.

3. Problems Faced by Students

- **Prolonged Resolution Delays:** Students frequently face long waiting times for basic Level 1 (L1) issues because tickets enter a general queue serviced sequentially by IT staff.
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- **Lack of After-Hours Support:** Standard helpdesk operations are tied to business hours. Issues occurring during late-night study sessions or weekends remain unaddressed until the next business day.
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- **Repetitive Information Entry:** When escalating an issue manually, students must re-explain problem details multiple times, leading to user fatigue and incomplete issue description entries.
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- **Unclear Status Tracking:** Students lack immediate feedback on whether their issue is a known common problem with a simple fix or requires manual technician intervention.

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4. Problems Faced by IT Support Staff

- **L1 Ticket Overload:** A disproportionate amount of IT staff time is spent answering repetitive, low-complexity queries (e.g., password resets, basic WiFi credentials) that could easily be self-resolved.

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- **Incomplete Initial Ticket Information:** Tickets submitted via simple forms or email frequently lack vital metadata (such as device type, exact error text, or specific troubleshooting steps already attempted). This forces staff into inefficient, back-and-forth communication.

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- **Manual Triage Inefficiencies:** Staff must manually categorize and route every incoming ticket, increasing operational overhead and delaying response times for high-priority infrastructure issues.

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5. Existing Process

1. **Issue Occurrence:** A student encounters a technical issue (e.g., WiFi failure, account lock).
- 2.
3. **Manual Ticket Creation / Email:** The student searches for a contact email or navigates to a static web form to lodge a ticket.
- 4.
5. **Queue Entry:** The request enters a centralized, un-triaged IT queue.
- 6.
7. **Information Gathering:** An IT support staff member reviews the ticket, finds it lacks diagnostic context, and emails the student for details.
- 8.
9. **Manual Resolution / Escalation:** Once details are provided, the staff member manually executes the resolution or re-assigns the ticket.
- 10.

6. Pain Points

- **High Ticket Volume:** Hundreds of identical requests hit the service desk weekly.

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- **Context Loss:** Information exchanged during initial troubleshooting is lost prior to formal ticket lodging.
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- **High Mean Time to Resolution (MTTR):** Simple requests take hours or days to resolve due to queue latency.
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- **Low Student Satisfaction:** The friction involved in resolving routine IT issues impacts the overall student experience.
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7. Root Causes

- **Absence of Automated Self-Service:** Existing portals do not offer interactive, guided decision trees to empower students to fix common issues independently.
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- **Lack of Integrated Conversational Workflows:** Ticket submission forms are completely decoupled from interactive troubleshooting tools.
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- **Static Support Infrastructure:** Knowledge resources exist as static, non-interactive documents that students rarely read.
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8. Effects of the Existing Process

- **Resource Misallocation:** Skilled IT engineers spend valuable hours on basic administrative and L1 tasks instead of maintaining core campus infrastructure.
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- **Academic Disruption:** Students encounter unnecessary roadblocks during critical academic tasks, assignments, and online examinations.
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- **Operational Bottlenecks:** Peak times (such as the start of a semester or exam weeks) lead to total support desk paralysis due to incoming request spikes.
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9. Problem Statement

"The current campus IT support infrastructure lacks an interactive, automated self-service mechanism for resolving routine technical issues."

Consequently, students experience significant resolution delays, and IT support staff are overwhelmed by high volumes of repetitive Level 1 tickets. There is an urgent need for a unified, intelligent portal that provides guided self-troubleshooting and seamless, context-aware ticket escalation directly within the ServiceNow platform."

10. Need for the Proposed System

To overcome these challenges, **CampusAssist AI** introduces an interactive Service Portal ([campusassist_home](#)) equipped with the **CampusCare AI** assistant widget ([campuscare_chatbot](#)). This system allows students to execute guided, self-paced troubleshooting across major IT categories (WiFi, Hardware, Software, Account Lock). When automated steps fail to resolve the problem, CampusAssist AI automatically transfers the diagnostic context to a native ServiceNow request form, creating a pre-filled record in the custom [x_2065601_campus_0_student_it_request](#) table.

11. Target Users

- **Students (End Users):** Primary consumers of the portal seeking instant resolution for IT problems.
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- **IT Support Staff (Fulfillers):** Fulfillers who receive well-structured, context-rich requests that require Level 2 or Level 3 technical intervention.
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- **System Administrators:** Administrators responsible for configuring and maintaining the ServiceNow platform, application scope, service portal widgets, and underlying tables.
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12. Project Motivation

The motivation behind CampusAssist AI is to showcase how enterprise Service Management platforms like ServiceNow can be leveraged to streamline campus operations. By automating L1 support through interactive chatbot technology and structured data routing via Glide APIs and URL parameters ([sysparm_query](#)), the project highlights modern best practices in IT Service Management (ITSM).

13. Expected Improvement

Metric / Dimension	Existing Campus Process	Proposed CampusAssist AI System
First Response Time	Hours (Subject to staff queues)	Instant (Automated via CampusCare AI)
L1 Ticket Deflection	0% (All queries become tickets)	High (30%–50% estimated deflection)
Diagnostic Context on Ticket	Low / Incomplete text entries	High (Structured troubleshooting log passed)
Service Availability	Business hours only	24/7 Portal Availability
User Experience	Fragmented, friction-heavy	Unified, interactive Service Portal

End of Document Metadata

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- **Purpose:** To define the campus IT support challenges, root causes, pain points, target user personas, and structural benefits of the proposed CampusAssist AI system.
- **Recommended Diagrams:** Problem-Context Diagram (Provided in Section 14).
- **Recommended Screenshots:** None required for this theoretical problem context document.